



Fig. 2: Circuit with only an inductor L

increase in voltage across R_C results in an equal decrease in voltage across T_1 . Therefore the output voltage from the amplifier, taken at the collector of T_1 with respect to the emitter, is a negative alternation of voltage that is larger than the input but has the same sine wave characteristics.

During the negative half of the input signal, the base current decreases, which results in a decrease in both emitter and collector currents. The decrease in current through R_C decreases its voltage drop and causes the voltage across the transistor to rise along with the output voltage. Therefore the output for the negative half of the input is a positive half of voltage that is larger than the input but has the same sine wave characteristics. By examining both input and output signals for one complete cycle of the input, we can see that the output of the amplifier is an exact reproduction of the input, except for the reversal in polarity and the increased amplitude.

To sum it up, the input signal is amplified because the small change in the base current causes a large change in the collector current by a factor of h_{fe} (β) approximately, and by placing the resistor R_C in series with the collector, voltage amplification is achieved.

Q3. Why does the current flowing through an inductor lag behind the voltage? I'd also like to get details on the instantaneous changes happening in the circuit, and the reason for a similar change in the case of a capacitor in the circuit.

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Via e-mail

A3. The following equation relates to v , i and L in a pure

inductor

$$v = L \frac{di}{dt} \quad (\text{By Faraday's law})$$

Voltage at any instant depends on the rate of change of inductor current at that instant but not on the value of current at that point. The voltage polarity is such that any current resulting from this voltage produces a flux that opposes the original change in flux. An inductor current cannot jump, it lags behind the voltage. In other words, in an inductor, building of current when voltage is applied, and collapsing of current when voltage is removed, will be opposed. In an inductor, energy $\frac{1}{2}LI^2$ is stored in the electromagnetic field.

Fig. 2 shows a circuit with only an inductor L.

Let the instantaneous current in the circuit be

$$i_L = I_m \sin \omega t \quad \dots\dots\dots (i)$$

This current produces a self induced emf e_L given by

$$e_L = -L \frac{di}{dt}$$

At any instant, two voltages are present in the circuit—one is the applied voltage v_L and the other is the induced voltage e_L .

$$\text{By KVL } v_L + e_L = 0$$

$$\begin{aligned} v_L &= -e_L = L \frac{di}{dt} = L \frac{d(I_m \sin \omega t)}{dt} \\ &= \omega L I_m \cos \omega t = \omega L I_m \sin (\omega t + 90^\circ) \quad \dots\dots\dots (ii) \end{aligned}$$

The comparison of Eqs. (i) and (ii) shows that for a purely inductive circuit the voltage leads the current by 90° or equivalently, the current lags behind the voltage by 90° .

Similarly, the relation between voltage, current and capacitance can be deduced; the formulae are the same, only the symbol differs. Where one has v , the other has i , and vice

versa; where one has the capacitance C , the other has L . It follows as the basic capacitor current-voltage relation is given by $i = Cdv/dt$ in place of $L di/dt$. Also, a purely capacitive current leads the voltage by 90° . In a capacitor, energy $\frac{1}{2}CV^2$ is stored in the electrostatic field.

Q4. Why the earth pin is larger in the 3-pin plug when compared to phase and neutral pins?

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A4. In an AC power plug, the earth pin is longer and of larger diameter as compared to the live and neutral pins. This arrangement performs three functions:

The longer earth pin provides a return path before the live pin engages. This way an appliance with a short to its case will simply trip a circuit breaker instead of electrocuting the user.

More the thickness of the conductor, less is the resistance. A thick earth pin provides a less resistant path for the current to be grounded. This is for the safety against shock. The leakage current, if any, will take the less resistant path through earth pin to ground rather than through the person operating the appliance.

Many wall sockets have shutters on the live and neutral lines to prevent children from inserting objects into the sockets such as conducting materials which may result in electric shock. Insertion of a longer pin helps in opening the shutters, facilitating the insertion of other two pins. For example, the third pin (larger diameter and length) of a mobile charger of a Nokia mobile phone has only one purpose, that is, opening of the shutter of the wall socket. It has no possibility of electric shock as the high voltage circuit is well inside the moulded case and the material of the pin is plastic.

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